

FSAE Design Spec Sheet

2025

Competitors: **Please read the Instructions** tips (tab below) prior to the completion and submission of this sheet.

Car No.	247
School	Gripton State University

Dimensions	Units				
Overall Dimensions	mm	Length:		Width:	
Wheelbase & Track	mm	Wheelbase:		Front Track:	
Center of Gravity Design Height	mm	CG Height:		Confirmed Via:	
Mass without driver	kg	Front:		Rear:	
Weight Distribution with 68kg driver		% Front:		% Left:	

Suspension Parameters	Units	Front			Rear		
Tire Size, Compound and Make							
Wheels (diameter, width)	inch	Diameter (col D): Width (col E):			Diameter (col G): Width (col H):		
Wheel material and construction							
Suspension Type							
Suspension design travel	mm	Jounce (col D): Rebound (col E):			Jounce (col G): Rebound (col H):		
Wheel rate (chassis to wheel center)	N/mm						
Roll rate (chassis to wheel center)	Nm/deg						
Sprung mass natural frequency	Hz						
Jounce Damping	% critical		at __ mm/sec:			at __ mm/sec:	
Rebound Damping	% critical		at __ mm/sec:			at __ mm/sec:	
Motion ratio	__.:1		Type:			Type:	
Ride Camber (Rate of Camber Change)	deg/m						
Roll Camber	deg/deg						
Static Toe (- out, + in)	deg						
Static camber	deg						
Static camber adjustment method							
Anti dive / Anti Squat	%						
Roll center height above ground, static	mm						
Roll center position at 1g lateral acc	mm	Height (col D): Lateral (col E):			Height (col G): Lateral (col H):		
Front Caster, Trail, and Scrub Radius		Caster (deg):		Kin Trail (mm):		Scrub Rad (mm)	
Front Kingpin Axis		Inclination (deg):		Offset (mm):			
Static Ackermann	%		Adjustable?				
Suspension Adjustment Methods							
Steer Ratio, C-Factor, Steer Arm Length		Steer Ratio (x:1)		c-factor (mm)		Steer Arm Length	

Brake System / Hub & Axle	Units	Front			Rear		
Rotors							
Master Cylinder							
Calipers							
Brake Pad/Lining Material							
Force and Pressures @ 1g Deceleration		Front Pres. (bar):		Rear Pres. (bar):		Pedal Force (kN)	
Upright Assembly							
Hub Bearings							
Axle type, size, and material							

Ergonomics	Units				
Driver Size Adjustments					
Seat (materials, padding/damping)					
Steering Wheel (dia, construction)		Diameter (mm)		Construction	
Shift Actuator (type, location)					
Clutch Actuator (type, location)					
Instrumentation					
Optional: Driver Safety Systems?					

Electrical	Units	
Power Management / Control		
Wiring / Loom / ECM mounting		
Battery / Charging System		
Grounding		
Driver Assist Systems		
Logging / Telemetry		
Special Sensing Technology		

Frame	Units	
Frame Construction		
Material		
Joining method and material		
Bare frame mass with brackets & paint	kg	Target: Physical Test:
Torsional stiffness	N-m/deg	Target: Simulated: Physical Test:
Torsional stiffness validation method		
Impact Attenuator configuration		
Impact Attenuator dimensions	mm	Width: Height: Depth:
Impact Attenuator energy capacity	kJ	Energy: Method:

Powertrain	Units	
Manufacturer / Model		
Cylinders & Fuel		Cylinders: Fuel Type:
Displacement & Compression		Displacement (cc): Compression (_:1):
Bore & Stroke	mm	Bore: Stroke:
Engine Output		Peak Power (kW) Peak Torque (Nm)
Design Speeds	rpm	Max Power: Max Torque: 80% Torque:
Induction (natural or forced, intercooled)		
Throttle Body / Mechanism		
Fuel Injection System (manf'r, and type)		
Fuel System Sensors (for fuel mapping)		
Fuel Pressure	bar	
Injector location		xx mm before _____ and directed towards _____
Intake Plenum		Volume (cc): Runner length (mm):
Exhaust Header Configuration		Effective Runner Length (mm): Variation (mm):
Exhaust Header Diameters		Primary (mm): Collector (mm):
Ignition System		
Ignition Timing		
Oiling System (wet/dry sump, mods)		
Engine Lubricants / Friction Treatment		
Coolant System and Radiator location		_____ mounted xx core _____ radiator , xxxx cfm fan mounted to _____
Fuel Tank Location, Type		Capacity (L):
Muffler		
Other significant engine modifications		

Drivetrain	Units	
Drive Type		
Differential System		
Final Drive Ratio	_:1	
Vehicle Speed @ max power (design) rpm	kph	1st gear: 2nd gear: 3rd gear:
Vehicle Speed @ max power (design) rpm	kph	4th gear: 5th gear: 6th gear:
Half shaft size and material		
Axle Joint type and grease used		

Aerodynamics (if applicable)	Units	
Type / Configuration		
Forces (at 80 kph, $\rho = 1.162 \text{ kg/m}^3$)		Downforce (N): % Front: Drag (N):
Coefficients & Reference Area		Cl: Ref. Area (m^2): Cd:
Noteable Features (active, etc)		

Other Information	Units	
Body Work (material, process)		
Optional Information		

Instructions / Tips for Design Specification Sheet

Formatting: **Please do not modify the formatting of the spec sheet.** It has been laid out to print on a single sheet (sided). While we acknowledge that there is little space in the cells, we ask that you be brief and include longer descriptions in the Design Report.

Completeness: Please complete the spec sheet in its entirety. "Unknown" is not an acceptable entry. The judges expect a value or description entered for each and every line item contained within. If by chance, the particular item does not apply to your design, please enter "N/A" in the space provided. N/A = Not Applicable.

Range of Values: In cases where you have yet to determine a specific value for a parameter, please input the range of values to fall within. Please make the noted range as tight (ie. narrow bandwidth) as you feel comfortable.

Units: For the majority of the specifications requiring a numerical input, the desired unit of measure has been provided. Please use the units as specified on this sheet as it makes it WAY easier on the judges. We have our reasons.

Comments / Clue Tabs: For many line items, there is additional descriptive information provided if you hover your cursor over the clue tab cell. Please use this information to guide your entries.

Understanding: Note that the specifications identified within the spec sheet contain the type of information the judge must understand, and be capable of discussing during the Design Event. Key factors during the discussion shall be the engineering rationale for each decision and resulting specification.

Consistency: All values contained within this spec sheet are expected to align with your Design report and your vehicle build. It will be expected that inconsistencies will be discussed during the Design event. Without adequate reasoning for the discrepancies, your design score may result.

Timeliness: Please submit the Spec Sheet and Design Report ON TIME. Late submission penalties can be significant. Points will be lost due to late submissions.

Data Entry Issues: If you are having issues with Excel not taking a value (especially if a negative sign is used) please use an " " as the first character.